

3.1 Analytical Question: The Centering Problem in Feature Space

Why we cannot directly subtract the mean in feature space

Kernel PCA works by implicitly mapping inputs $x \in \mathbb{R}^D$ to a (possibly infinite-dimensional) feature space F via $\Phi : \mathbb{R}^D \rightarrow F$. The kernel trick lets us compute inner products $\langle \Phi(x), \Phi(y) \rangle = k(x, y)$ without ever computing $\Phi(x)$ explicitly. Because $\Phi(x)$ may be very high- or infinite-dimensional (e.g. RBF kernel), we cannot form or subtract the mean vector $\mu_\Phi = \frac{1}{N} \sum_{i=1}^N \Phi(x_i)$ explicitly. Any centering operation must therefore be written using only kernel evaluations $k(x_i, x_j)$.

Derivation of the centered kernel matrix

Let $\tilde{\Phi}(x_i) = \Phi(x_i) - \mu_\Phi$. The centered Gram (kernel) matrix has entries

$$K_{ij}^{(c)} = \langle \tilde{\Phi}(x_i), \tilde{\Phi}(x_j) \rangle = \langle \Phi(x_i), \Phi(x_j) \rangle - \langle \Phi(x_i), \mu_\Phi \rangle - \langle \mu_\Phi, \Phi(x_j) \rangle + \langle \mu_\Phi, \mu_\Phi \rangle.$$

Write each term using the kernel $k(\cdot, \cdot)$:

$$\begin{aligned}\langle \Phi(x_i), \mu_\Phi \rangle &= \frac{1}{N} \sum_{k=1}^N k(x_i, x_k), \\ \langle \mu_\Phi, \Phi(x_j) \rangle &= \frac{1}{N} \sum_{k=1}^N k(x_k, x_j), \\ \langle \mu_\Phi, \mu_\Phi \rangle &= \frac{1}{N^2} \sum_{k=1}^N \sum_{\ell=1}^N k(x_k, x_\ell).\end{aligned}$$

So elementwise

$$K_{ij}^{(c)} = K_{ij} - \frac{1}{N} \sum_k K_{ik} - \frac{1}{N} \sum_k K_{kj} + \frac{1}{N^2} \sum_{k,\ell} K_{k\ell}.$$

Matrix form

Let $\mathbf{K}$ be the $N \times N$ Gram matrix with entries $K_{ij} = k(x_i, x_j)$. Let $\mathbf{1}_N$ denote the $N \times N$ matrix with every entry $1/N$, and let

$$\mathbf{H} = \mathbf{I} - \mathbf{1}_N = \mathbf{I} - \frac{1}{N} \mathbf{1} \mathbf{1}^\top$$

be the centering (projection) matrix. Then the centered kernel matrix is compactly

$$\mathbf{K}^{(c)} = \mathbf{H} \mathbf{K} \mathbf{H}$$

which is algebraically equivalent to the elementwise formula above. Geometrically, $\mathbf{H}$ projects out the component along the mean direction in feature space.

3.2 Numerical Problem (corrected)

Data and kernel

$$X = [-2, -1, 1, 2], \quad k(x, y) = (xy + 1)^2.$$

(a) Gram matrix $\mathbf{K}$

Compute $K_{ij} = k(x_i, x_j)$. The resulting Gram matrix is

$$\mathbf{K} = \begin{pmatrix} 25 & 9 & 1 & 9 \\ 9 & 4 & 0 & 1 \\ 1 & 0 & 4 & 9 \\ 9 & 1 & 9 & 25 \end{pmatrix}.$$

(Checks: row sums are $[44, 14, 14, 44]$ and total sum 116.)

(b) Centered kernel matrix $\mathbf{K}^{(c)}$

Using $\mathbf{K}^{(c)} = \mathbf{H}\mathbf{K}\mathbf{H}$ (equivalently the elementwise formula), we get

$$\mathbf{K}^{(c)} = \begin{pmatrix} 10.25 & 1.75 & -6.25 & -5.75 \\ 1.75 & 4.25 & 0.25 & -6.25 \\ -6.25 & 0.25 & 4.25 & 1.75 \\ -5.75 & -6.25 & 1.75 & 10.25 \end{pmatrix}$$

(If you compare with your intermediate arithmetic, note a prior sign/arithmetic slip produced different small numbers — the display above is the correct centered matrix.)

(c) Eigen-decomposition (principal components in feature space)

Compute eigenvalues/eigenvectors of $\mathbf{K}^{(c)}$. The (rounded) eigenvalues are

$$\lambda \approx [0, 0, 9, 20],$$

i.e. two numerically-zero eigenvalues and two positive eigenvalues 9 and 20. (Zero eigenvalues appear because centering and kernel structure reduce effective rank.)

The largest eigenvalue is

$$\lambda_{\max} = 20$$

with a corresponding (unit) eigenvector that can be written up to sign as

$$\mathbf{v}_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} -2 \\ -1 \\ 1 \\ 2 \end{pmatrix}.$$

(Indeed $\mathbf{v}_1$ is proportional to the original inputs $[-2, -1, 1, 2]^\top$. The sign of an eigenvector is arbitrary.)

The second nonzero eigenvalue $\lambda_2 = 9$ has eigenvector (up to sign)

$$\mathbf{v}_2 = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix},$$

which captures the orthogonal mode of variation.

Interpretation. The top eigenvector separates points by *magnitude and sign* (large negative vs large positive), which is intuitively the main variation in this symmetric dataset under the chosen polynomial mapping.